MNIST Handwritten Digit Classification using CNN

Objective:
To build a Convolutional Neural Network (CNN) model to classify handwritten digits using the MNIST dataset.

Dataset:
The MNIST dataset consists of 28×28 grayscale images of handwritten digits (0–9).

Model Architecture:
The model includes convolutional layers, max pooling layers, a flatten layer, and dense layers for classification.

Training:
The model was trained for 5 epochs using Adam optimizer and categorical crossentropy loss function.

```
history = model.fit(x_train, y_train,
                    epochs=5,
                    validation_data=(x_test, y_test))
```

```
Epoch 1/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 54s 28ms/step - accuracy: 0.9396 - loss: 0.2011 - val_accuracy: 0.9854 - val_loss: 0.0424
Epoch 2/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 52s 28ms/step - accuracy: 0.9784 - loss: 0.0743 - val_accuracy: 0.9884 - val_loss: 0.0348
Epoch 3/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 52s 28ms/step - accuracy: 0.9828 - loss: 0.0546 - val_accuracy: 0.9900 - val_loss: 0.0297
Epoch 4/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 51s 27ms/step - accuracy: 0.9871 - loss: 0.0432 - val_accuracy: 0.9913 - val_loss: 0.0259
Epoch 5/5
1875/1875 ━━━━━━━━━━━━━━━━━━━━ 52s 27ms/step - accuracy: 0.9884 - loss: 0.0374 - val_accuracy: 0.9915 - val_loss: 0.0260
```

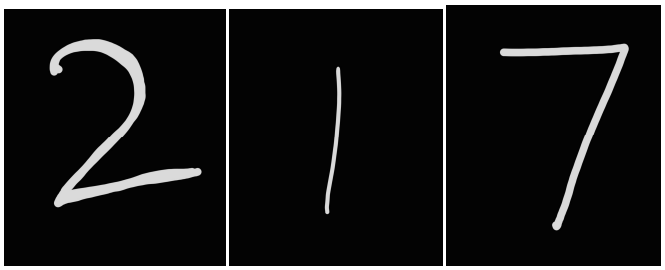
Results:
The model achieved a test accuracy of approximately **99.15%**.

```
test_loss, test_acc = model.evaluate(x_test, y_test)
print("Test Accuracy:", test_acc)
```

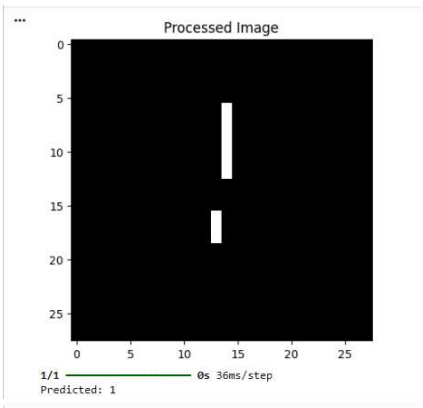
```
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 8ms/step - accuracy: 0.9915 - loss: 0.0260
Test Accuracy: 0.9915000200271606
```

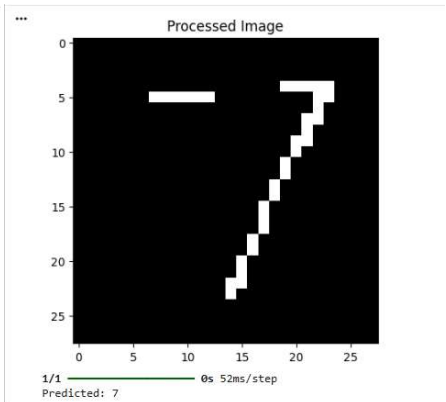
Custom Image Testing:
The model was tested on custom handwritten digits (2, 7, 1), and predictions were correct.

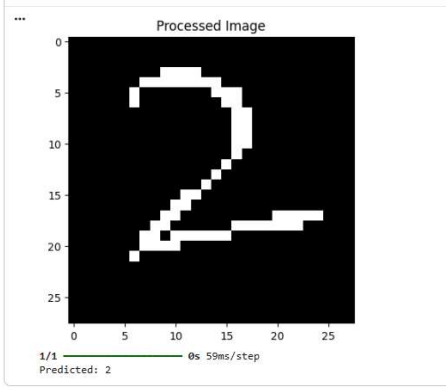
Custom images given :



Predicted outputs :







Conclusion:
The CNN model performed effectively and accurately classified handwritten digits.